

Count significant digits



1 State the number of significant figures in the following numbers:

- | | | | |
|------------------|----------------------|-------------------|-----------------|
| a 108 486 | b 5105 | c 0.20 | d 10.20 |
| e 84.00 | f 0.007 193 0 | g 0.004 58 | h 2.0058 |

2 State the smallest possible number of significant figures in the following numbers:

- | | | | |
|--------------------|-----------------|--------------------|---------------------|
| a 1 500 000 | b 26 500 | c 1 060 000 | d 82 420 000 |
|--------------------|-----------------|--------------------|---------------------|

3 A rainfall gauge has markings every 10 mL. If 369 mL of rain fell in this gauge, to how many significant figures would the rain be recorded?

4 Beth just bought a house for \$858 400. When asked for the price by her friend, she said she paid around \$860 000. To how many significant figures did Beth round the price in her answer?

5 A number has been rounded to two significant figures. If the rounded number is 4500, find the largest possible value of the original number.

Round with significant figures

6 Consider the number 1083.

- a** Round 1083 to:
- | | |
|---------------------------------|-----------------------------------|
| i One significant figure | ii Two significant figures |
|---------------------------------|-----------------------------------|
- b** Find the difference between these two rounded values.

7 Round the following numbers to two significant figures:

- | | | | |
|-----------------|-----------------|---------------|------------------------|
| a 2584 | b 10 694 | c 9249 | d 215.99 |
| e 0.7218 | f 0.0297 | g 4.95 | h 0.006 037 736 |

8 Round the following numbers to three significant figures:

- | | | | |
|---------------|------------------|-----------------|--------------------|
| a 1771 | b 461 585 | c 6484 | d 20 994 |
| e 9248 | f 539.99 | g 0.5628 | h 0.073 582 |

9 Round the following numbers to four significant figures:



a 801 600 713

b 0.060 070 047

c 11.8483

d 2.159 76

e 4.504 13

f 1.599 64

g 90.0335

h 0.598 71

i 41.5005

j 9.9996

k 0.093 845

l 808.5042

10 Round the following numbers to five significant figures:

a 5.659 64

b 5.600 96

c 12.006 65

d 39.999 95

e 7.034 500

f 0.041 454 9

g 99.097 906

h 9.999 99

11 Round 7231.0108 to five significant figures.

12 Use your calculator to round the following to three significant figures:

$$\frac{(9.45)^9 + 8\pi}{\sqrt{\left(\frac{3}{2} + 5^2\right)}}$$